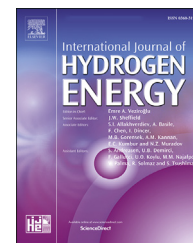


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GIS based multi-criteria decision making for solar hydrogen production sites selection in Algeria

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HIGHLIGHTS

- The paper addresses the problem of site selection for solar hydrogen projects.
- Land suitability index is computed to determine the best sites.
- MCDM coupled with GIS is a powerful tool for effective evaluation of the solar hydrogen production sites selection.

ARTICLE INFO

Article history:

Received 7 May 2019

Received in revised form

9 September 2019

Accepted 12 October 2019

Available online 9 November 2019

Keywords:

Hydrogen energy

Geographical information system (GIS)

Multi-criteria decision making (MCDM)

Site selection

Solar energy

Algeria

ABSTRACT

Hydrogen production from renewable energy sources appears to be an interesting solution for reducing greenhouse gas emissions and ensuring the energy security supply. This paper develops an integrated framework to evaluate land suitability for hydrogen production from solar energy site selection that combines multi-criteria decision making (MCDM) with geographical information systems (GIS); an application of the proposed framework for Algerian country. In GIS two types of criteria will be taken: constraints and weighting criteria. Constraints criteria will make it possible to reduce the area of study by discarding those areas that prevent the implementation of installing solar hydrogen production systems. These criteria will be obtained from the legislation (land use, water bodies, waterways, roads, railways, power lines, and also their buffer around them). Weighting criteria will be chosen according to the objective to be reached, in this case they will be the hydrogen demand, potential solar hydrogen production, digital elevation models (DEMs), slope, proximity to roads, railways, and power lines. Through the use of MCDM the criteria mentioned will be weighted in order to evaluate potential sites to locate a solar hydrogen production installation system. Analysis and calculation of the weights of these criteria will be conducted using Analytic Hierarchy Process (AHP). As a result, the final index model was grouped into four categories as “very low suitability”, “low suitability”, “moderate suitability” and “high suitability” with a manual interval classification method. The results indicate that 10.34% (246,272.02 km²), of the study area has very low suitability, 60.75% (1,446,907.65 km²) has low suitability, 6.68% (159,100.3 km²) has moderate suitability and 0.49% (11,669.21 km²) has high suitability for a solar-powered hydrogen production installation system. The other 21.74% (517,790.5 km²) of the study area is not suitable for such projects. The sensitivity analysis highlights that the suitable sites for solar hydrogen production installation system are dependent on the weights of the criteria that influence

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<https://doi.org/10.1016/j.ijhydene.2019.10.099>

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the decision. The MCDM methodology integrated with GIS is a powerful tool for effective evaluation of the solar-powered hydrogen production sites selection.

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Introduction

The energy demand is increasing around the world, which is driven mainly by population growth; more than 80% of today's energy demand is produced using fossil fuels [1]. The burning of fossil fuels is mainly the largest source of emissions of carbon dioxide by humans, which is one of the greenhouse gases that contribute to global warming. According to the International Energy Agency, the world CO₂ emissions from fuel combustion rose 108% from 1973 to 2015 [2]. In 2014, by economic sector, the transportation accounts for a significant share of the global fossil fuel combustion-related CO₂ emissions, which is responsible for over 23%. Almost all (95%) of the world's transportation energy comes from petroleum-based fuels, largely gasoline and diesel. The world must therefore balance the role of energy in social and economic development with the need to decarbonize, reduce our reliance on fossil fuels, and making transition towards lower-carbon energy sources.

The latest estimate of OPEC, published in 2016, Algeria is estimated at 12.2 billion barrels of proven reserves of conventional crude oil, and proven reserves of natural gas from Algeria were estimated at about 4.5 trillion cubic meters placing the country 7th in the world [3]. In 2017, the overall primary production of hydrocarbons in Algeria, all end products included, reached 196.5 Mtoe. Crude oil production was approximately 49.3 Mtons. NG production has reached 135 milliard m³. The condensate production leveled 14 Mtons and liquid petroleum gases (LPG) primary production attained 14 Mtons [4]. The total installed power capacity and electricity generation in Algeria has a value of 15.1 GW at the end of 2013, up from 12.9 GW at the end of 2012 and 11.4 GW at the end of 2011. The majority of electrical generation capacities (51%) comes from open cycle gas turbines, 29% combined cycle gas turbines, 16% conventional steam turbines and 2% diesel oil, 1% hydraulic and 1% hybrid solar/gas [5]. Most of Algeria's population is connected to the National Grid (99%). Total national energy consumption reached 59.6 Mtoe in 2017, up +2.1% from 2016. It accounts for more than one third (35.9%) of total production [6]. By sector, residential and other sector accounted for 44.4% of total energy consumption followed by the transport sector with 33.3% and industrial with 22.3% [6]. The consumption of the transport sector reached 14.89 Mtoe in 2017 [6]. The fuel consumption for road transport mode is 95% of the sector's consumption (increase in the national car fleet, with nearly 192,171 new vehicles imported in 2016) [7].

There are many studies in the literature that have been widely discussed the vast promise of hydrogen as a future solution to address environmental and energy security problems posed by current transportation fuels, which that must be defined in the coming years [8]. A large number of

hydrogen-fueled vehicle projects have been implemented in recent years by manufacturers such as Toyota Motor and Nissan Motor have begun producing hydrogen-fueled vehicles [9]. The Mirai, the fuel cell vehicle manufactured by Toyota, has already been mass-produced in Japan, the U.S and also some European countries [10]. In terms of selling figures, according to a report by market research from Ref. [11], 6,364 hydrogen fuel cell vehicles have been sold in the last of 2017. Moreover, the future view is conveyed as Ref. [12]; there is not at least under 400 million cars driven by hydrogen in 2050. However, the transition from fossil-fueled vehicles to environmentally friendly hydrogen-fueled vehicles depends on the availability of technologies to produce hydrogen from cleaner, renewable, and sustainable sources as well as the wide availability of hydrogen refueling stations [13].

Hydrogen can be produced from renewable energy sources through a variety of pathways and methods [14]. Additionally, there are many studies related to the hydrogen production from solar energy. Boudries [15] evaluated the hydrogen production in different sites in Algeria from electrolyzer-concentrating photovoltaic system. He conducted a comparison between two concentrating technologies: Fresnel mirrors-photovoltaic concentrator and a parabolic trough-photovoltaic concentrator, and his results showed that the average hydrogen production during the low irradiated month is around 0.14 kg/m²/day per unit cell area when using Fresnel mirrors concentrator and 0.10 kg/m²/day using the parabolic trough concentrator. This production reaches 0.19 and 0.17 kg/m²/day during the highly irradiated month by the two selected technologies respectively. Saadi et al. [16] studied the hydrogen production from solar energy available in Biskra region in the south-east of Algeria by the Proton Exchange Membrane (PEM) electrolyzer. Their results showed that Biskra area has the great perspective to produce solar hydrogen not only in Algeria but also in North Africa contrities. In addition, other researches were carried out on different countries and regions, e.g., Saudi Arabia [17], Egypt [18], Algeria [19], Turkey [20], Morocco [21], Pakistan [22], South Africa [23], Jordan [24], Iran [25]. The main objective of these studies was to make the hydrogen production the focus of policy makers and investors' attention.

There are some common methods of producing hydrogen from water such as thermolysis, radiolysis, photocatalytic water splitting, electrolysis, thermochemical cycle, ferrosilicon method and etc [26,27]. But the use of renewable energy sources for hydrogen production by water electrolysis is advantageous because it reduces environmental pollution [28]. Scamman et al. [29] investigated the performance of a combined PEM electrolyzer and PV plant at three different climates. They estimated hydrogen yield at different geographical locations that was in range of 25–65 kg p.a. for a

1.6 kW electrolyzer with fixed-tilt PV panels. Ankica et al. [30] presented a system for hydrogen production via water electrolysis using a 960 Wp solar power plant. The calculated hydrogen rate of production was 1.138 g per hour. During their study the system produced 1.234 MWh of energy, with calculated of 1.31 MWh, which could power 122 houses, and has offset 906 kg of carbon or an equivalent of 23 trees.

The integration of on-site hydrogen production with renewable energy technologies such as solar energy systems offer a practical means of supplying hydrogen on a small scale compared with central hydrogen production stations, which require significant capital investment to build a reliable hydrogen transport and delivery infrastructure [31,32]. To promote the development of the infrastructure for renewable hydrogen production, it is essential to find suitable sites for such development. Land-use suitability analysis is a tool used to identify the most suitable places for locating future land uses according to specific requirements, preferences, or predictors of some activity [32]. Determining suitable land for a particular use is a complex process involving multiple aspects that may relate to biophysical, socio-economic and technical aspects [33].

The choice of an installation site from other sites is MCDM problem containing many criteria [34]. The MCDM includes several different methods [35], e.g., TOPSIS [36], ELECTRE III [37], AHP [38] ... etc. Pohekar et al. [39] presented a literature review of more than 90 published papers on multi-criteria decision making on sustainable energy planning and observed that the analytic hierarchy process (AHP) is the most popular technique. AHP is a systematic approach developed in the 1970s by Thomas Saaty to deal with decision making problems in complex and multi-criteria situations [40]. AHP allows the combination of different evaluation criteria.

Additionally, AHP permits the participation of different stakeholders of energy related fields, by the implementation of pair-wise comparisons of evaluation criteria, for their relative importance determination. AHP method has been used successfully in various fields [35].

The integration of multi-criteria methods in a geographical information system has the potential to enhance its analytical strength. There are several studies on this topic to identify the suitability of sites for solar energy [38,41–48], wind energy [49,50], or hybrid system [51,52]. One common point of these research studies is the use of the geographical information system coupled with multi-criteria decision making to select the suitable locations for solar and/or wind energy. In their studies, authors have taken into account constraints criteria and different weighted criteria. Constraints criteria that may prevent the use of solar and/or wind energy on such territory, like the land use, water bodies, waterways, roads, railways, power lines, and also their buffer around these zones. Weighting criteria will be chosen according to the objective to be reached. Authors agreed that the MCDM methodology integrated with GIS is a powerful tool to select suitable sites for renewable energy. Ghasempour et al. [53] reviewed more than 40 published articles employing MCDM approach in selecting site location for solar energy and selecting the best technology. They concluded that the main criteria in site selection are economy, environment, risk, geography, vision, ecology, society, and climate, and also each of the mentioned criteria has a number of sub-criteria. For selecting solar technology, many criteria affect decision making such as technical aspects, economical aspects, feasibility, efficiency, land usage, flexibility, CO emission, reliability and accuracy. Moreover, they concluded that solar radiation is one of the most important factors since it is mentioned in all studies for site selection.

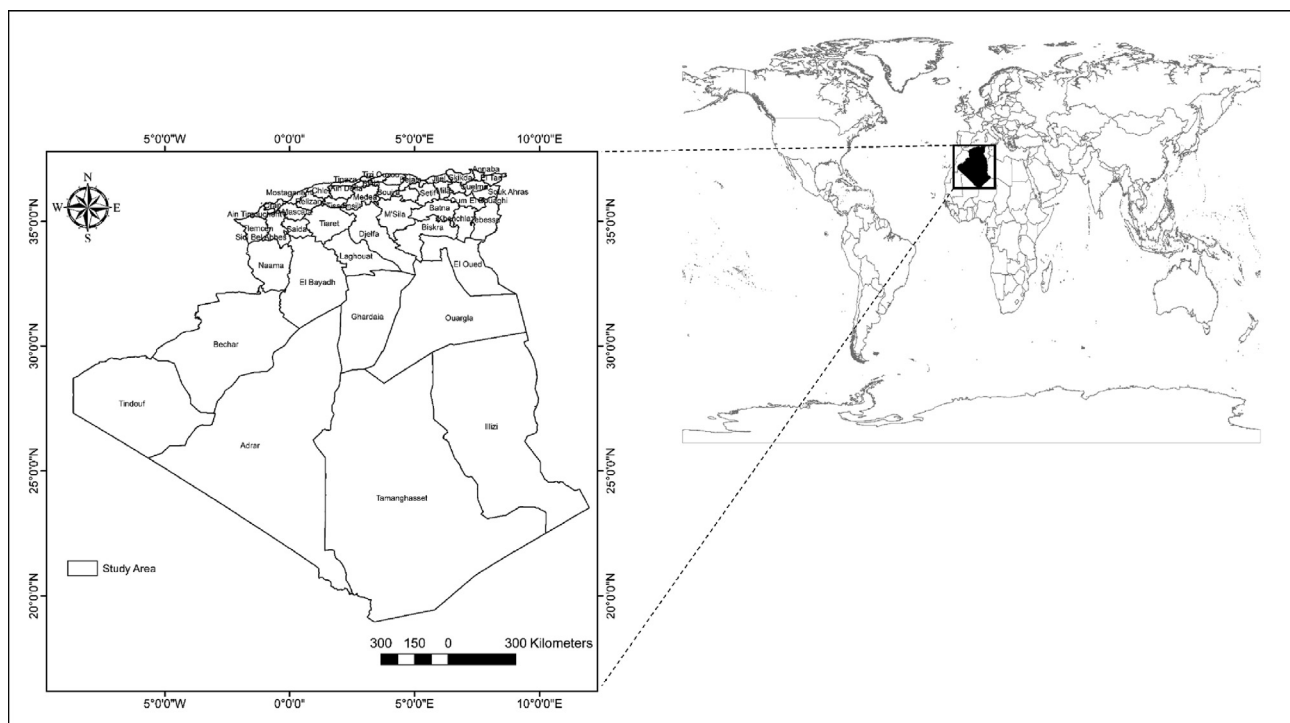


Fig. 1 – Location map of the study area.

As regards the analysis of suitable sites for installing renewable hydrogen production systems, Rezaei-Shouroki et al. [54] used Data Envelopment Analysis (DEA), AHP and Fuzzy Technique for Order of Preference by Similarity to Ideal Solution (FTOPSIS) methods to prioritize of wind farm locations for hydrogen production in Fars province, Iran. Dagdougui et al. [55] proposed a GIS-based decision making methodology for selecting the most promising locations for installing renewable hydrogen production systems in Liguria region, Italy.

In a previous research the potential for producing hydrogen was analyzed from solar, wind and geothermal resources in Algeria. Gouareh et al. [56] were used GIS based methodology combined with several criteria to analyze the spatial distribution of CO₂ emission sources, and to identify the most suitable locations for integrating geothermal heat extraction processes to evaluate the potential for both electric and hydrogen production in Algeria. Rahmouni et al. [57] estimated the potential for producing hydrogen from both solar photovoltaic and wind resources in Algeria. Their objective was to analyze renewable resource data both statistically and graphically using GIS and to evaluate the

availability of renewable electricity production potential from these key renewable resources in order to produce of hydrogen via the electrolysis process. Finally, they compared the results of the estimated hydrogen potential from solar and wind resources for each region. Messaoudi et al. [58] developed a methodology for site selection of hydrogen refueling stations with on-site hydrogen production from wind energy sources. The authors examined a case study for the region of Adrar in Algeria and they followed an analytical hierarchy process belonging to the multi-criteria decision making tools in order to identify which of the available petrol stations can be reverted to H₂ refueling stations. Out of 24 conventional petrol stations in total, 15 were investigated and only 4 sites were found suitable for wind powered onsite H₂ stations. The rest 11 however, could be modified as off-site hydrogen stations in the retail refueling market.

The aim of this study was to produce a decision support system (DSS) which can assist authorities and decision makers to identify priority sites of solar resources to produce hydrogen energy for transport sector in Algeria, using multi-criteria coupled with geographic information systems. To date, this is the first detailed study to explore Algerian country

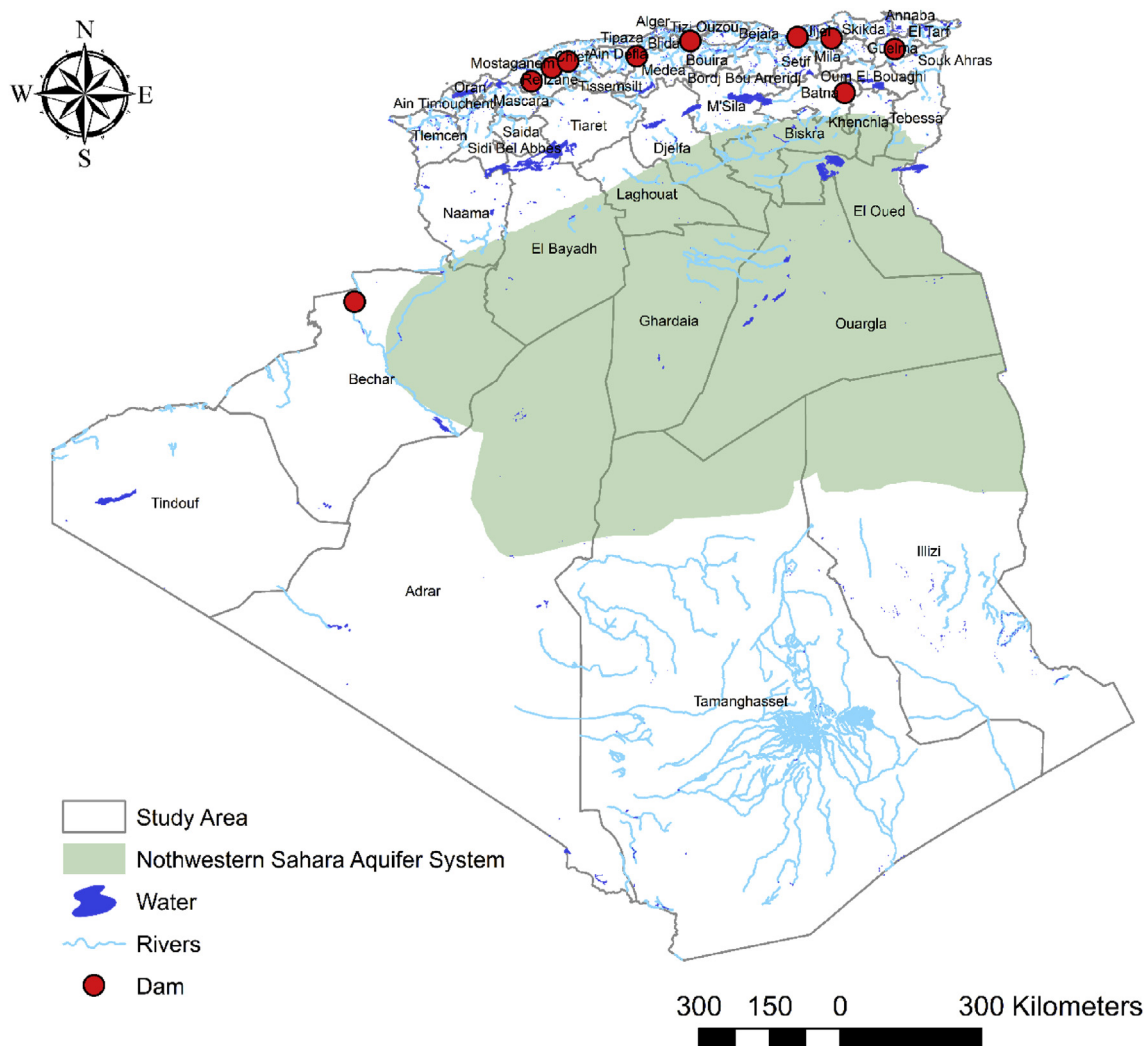


Fig. 2 – Water resources in Algeria.

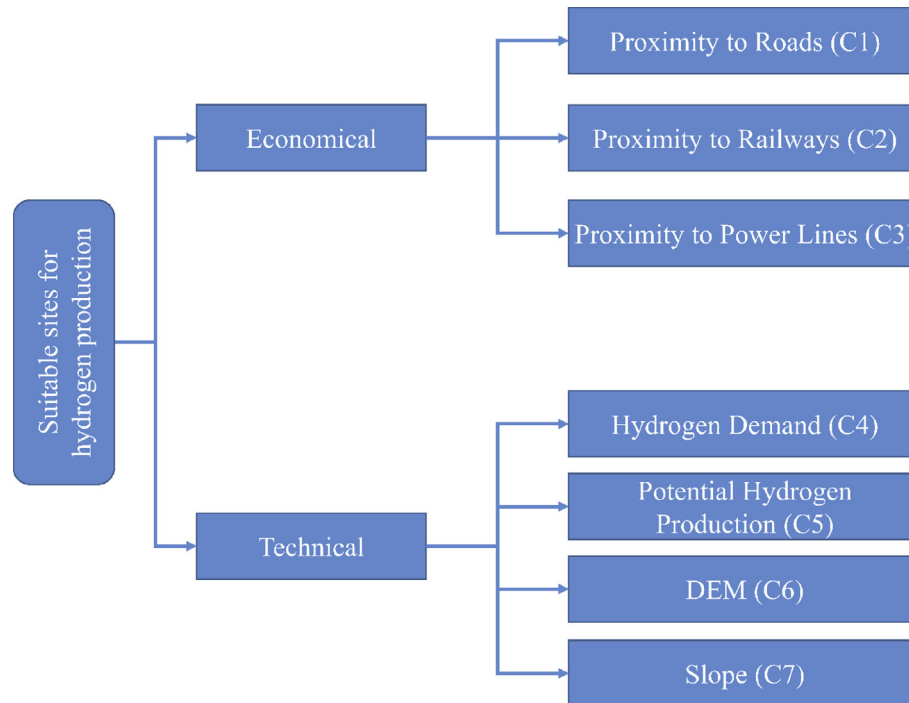


Fig. 3 – Decision criteria considered in renewable hydrogen production site selection.

with a view to identifying potential locations for siting solar hydrogen production sites simultaneously using AHP and GIS.

The structure of this paper is as follows: In Section [Study area](#), study area description of Algeria is illustrated. In Section [Energy context in Algeria](#), a brief overview of energy context in Algeria will be presented. In Section [Methodology](#), methodology is discussed. In section [Results and discussion](#) deals with the results and, lastly, in the final Section, we provide the main conclusions and suggest future works.

Study area

Algeria is situated in northern Africa, bordered to the east by Tunisia and Libya, to the southeast by Niger, to the southwest by Mali, and to the West by Mauritania, occidental Sahara, and Morocco as shown in [Fig. 1](#). Algeria is the largest country in Africa with a total surface of 2,381,741 km², with 1200 km of coastline extends to the Mediterranean Sea which accounts for barely 4% of the territory, is home to the majority of the population and economic activities. The South is representing nearly 87% of the country's surface area, remain underpopulated (9% of the population) and poorly endowed in terms of economic and social infrastructure. According to the National Office of Statistics (ONS) in 2017, the population is slightly over 41.2 million inhabitants, where the population density in the north is important (2511 Pers./km² in Algiers) than in the south (0.3 Pers./km² in Tindouf). Northern Algeria lies within the temperate zone, and its climate is similar to that of other Mediterranean countries, despite the fact that the decent variety of the alleviation gives sharp complexities in

temperature. The coastal region has a pleasant climate, with winter temperatures averaging from 10° to 12 °C and average summer temperatures ranging from 24° to 26 °C. Further inland, the climate changes; winters average 4° to 6 °C, and summers average 26° to 28 °C. In the Sahara Desert, temperatures range from 10° to 34 °C, with extreme highs of 49 °C.

Energy context in Algeria

Solar energy projects

According to satellites' evaluation of the German Aerospace Center (DLR), Algeria has one of the highest solar potentials in the entire Mediterranean region, with about 170 TWh per year for thermal solar and more than 13.9 TWh per year for photovoltaic [59]. In 2016, the country's total solar installed capacity was more than 240 MW. Under the National Development Plan for Renewable Energies, the country aims to install renewable generation capacity of 22 GW by 2030, of

Table 1 – GIS map layers used in the study.

Data	Scale	Data source
Roads	1:13,906,327	INCT, OSM, and Google earth pro
Railways	1:13,909,017	INCT, OSM, and Google earth pro
Power lines	1:14,032,441	SONELGAZ
DEM	1:15,114,688	CGIAR-CSI
Waterways	1:14,563,405	INCT, OSM, and Google earth pro
Water bodies	1:13,817,872	INCT, OSM, and Google earth pro
Land use	1:14,158,497	INCT, OSM, and Google earth pro

which 12 GW will be intended to meet the domestic electricity demand and, under certain conditions, 10 GW destined for export, with solar accounting for almost 60% of the capacity [60]. One of the most important projects is that located in Hassi R'Mel in northern Algeria, where Sonatrach (state-owned company for exploration, transport and marketing of petroleum, natural gas and related products), and Sonelgaz (state-owned company for electricity generation, transmission, distribution and control) formed a new, renewable energy joint venture company, called New Energy Algeria (NEAL). The complex comprise a 130 MW combined cycle, with a gas turbine power of the order of 80 MW and a 75 MW steam turbine [61]. Solar electricity devices are often referred to as PV or concentrating solar power (CSP). PV is considered as one of the key technologies that are at the heart of the energy technology revolution, because they can make the largest contributions to reduce greenhouse gas emissions [62].

Hydrogen production projects

In 2005, the role of hydrogen economy in Maghreb has been discussed by within the first international workshop on hydrogen economy “WIH2'05” which was held in Algiers, many recommendations were suggested developing hydrogen economy by research. The major recommendations are that

creation of The Algerian Hydrogen Association (A2H2), and the launch of a large Maghreb-Europe cooperation project [59,61,63].

Water resources

Algeria is country rich in natural resources, Algeria has a high variation of water potentialities. Considering that the Sahara Desert covers most of Algeria (87%), where precipitations are quasi-null, but which conceals important fossil underground water resources, whereas the surface water represent 0.37 billion m^3 and the underground resources represent 5 billion m^3 that provide the country with approximately 66% of its annual requirement [64,65]. The Northern part of the country is characterized by a Mediterranean climate, with renewable surface and underground water resources which respectively represents 7.4 and 2.6 billion m^3 [65]. The surface water is located in the north area which covers about 7% of the territory. And the rest came from seawater desalination with overall production capacity around 2,310,000 m^3 per day [65]. The North Western Sahara Aquifer System [NWSAS] are jointly exploited with Libya and Tunisia. It consists of two major superimposition aquifers layers which are the Terminal Complex (CT) and the Continental Interlayer (CI) as shown in Fig. 2.

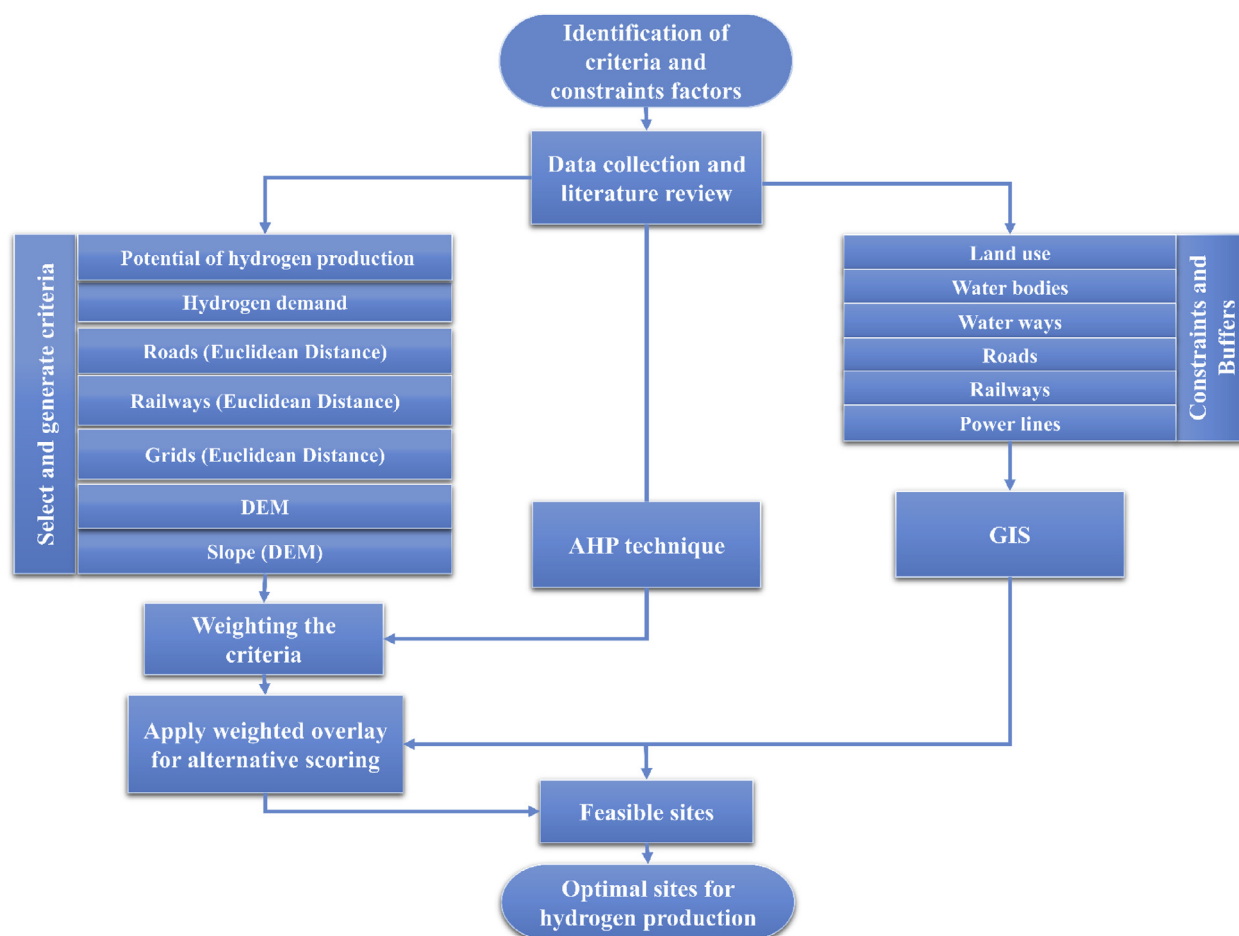


Fig. 4 – Flowchart of the methodology.

Table 2 – Data and information and defined criteria for unsuitable sites for hydrogen production installation system powered by solar energy.

Category	Data	Criteria (unsuitable land)	References
Economical Criterion	Roads	Area in distance < 500 m	[71]
	Railways	Area in distance < 500 m	[55]
	Electric power line	Area in distance < 250 m	[55]
Technical Criterion	DEM	Area with elevation of >2000 m	[72]
	Slope	Area have slope angles greater than 5.71°	[73]
Restriction	Water Bodies and Waterways	Area in distance from waterway < 500 m	[74]
		Area in distance from water bodies < 1000 m	[75]
	land use	Area in distance < 500 m	[76]
			[55]
			[73]

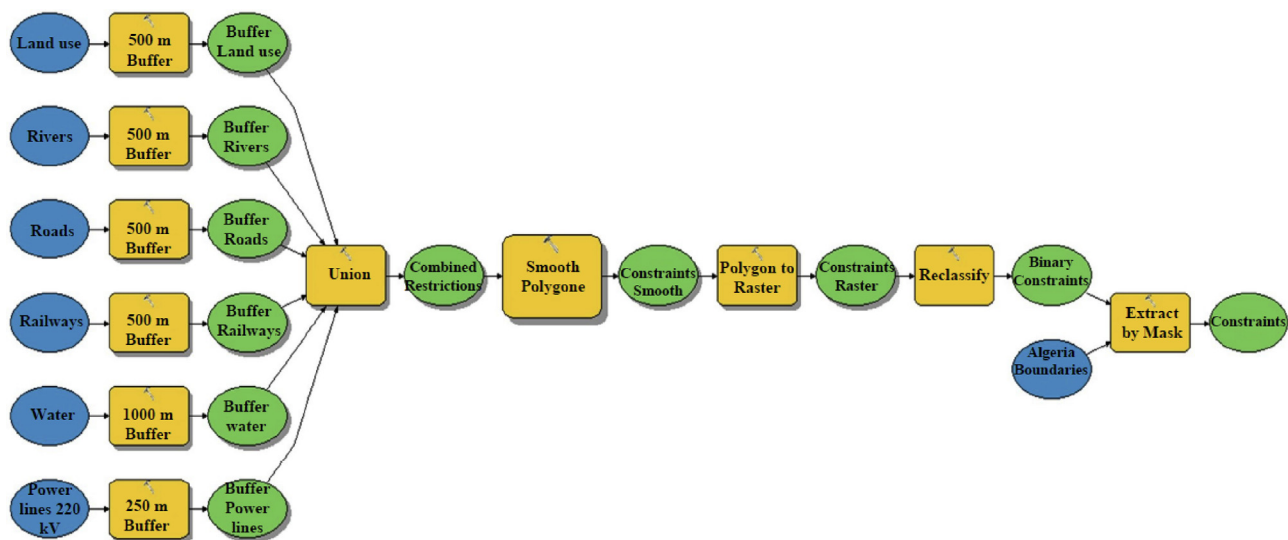
Methodology

Based on several literatures, case studies concerning solar PV power plant site selection and experts' opinion in the field of energy, seven criteria were selected for evaluating of suitable sites for solar hydrogen production in Algeria [38,43,55,66–69]. Fig. 3 presents the adopted criteria, which are divided generally into two groups: economical criteria which include proximity to roads, proximity to railways, and proximity to power lines and technical criteria which include hydrogen demand which contains the population, potential renewable hydrogen production, DEM, and slope.

Additionally, land use, waterways, water bodies, and also a buffer zone around the roads, railways, waterways, water bodies, land use, power lines, the area that greater than 2000 m elevation and the slope greater than 5.71° have been imposed as constraints. The data were collected from different sources including governmental agencies, open sources, and related literature as shown in Table 1.

A five phase analysis is performed to simplify decision support for renewable hydrogen production sites selection (see Fig. 4).

- In the first phase, a GIS map overlay technique is applied take different constraints and restrictions into consideration to rule out unsuitable sites;
- In the second phase, a reclassification technique is performed for each criterion in order to evaluate its particular suitability for renewable hydrogen production;
- In the third phase, an AHP technique is applied to determine the relative importance and priority weight of each criterion;
- In the fourth phase, the overall evaluation of the candidate sites is determined by applying a raster calculator approach in the ArcGIS tool, the main concept of this technique is that it allows you to create and execute Map Algebra expressions in a tool;
- In the final phase, the unfeasible sites generated in the first phase are excluded from potential areas for the selection of renewable hydrogen production sites. By doing so, all

**Fig. 5 – Restrictions part of the model.**

excluded areas that received a value of 0 keep that value, while areas that obtained a value of 1 in the first phase adopt the values calculated in the second phase (reclassified from 1 to 10 with the highest score being more suitable) [66].

Exclusion area

The first step of this analysis is the exclusion of areas which are existing in fact whether with lawful authority or not are unsuitable for locating renewable hydrogen production installation system. This contains areas such as land use, water bodies, waterways, roads, railways, power lines, and also their buffer around these zones, i.e. minimum distances, around those areas are excluded. and eliminate areas that have high elevation land ($>2000\text{ m}$) and higher slope lands ($>5.71^\circ$). Table 2 provides an overview of the excluded areas and their buffer. These constraints were used in many literature studies for suitable site selection in different fields [38,66,70].

The raw data were collected from different sources including governmental agencies, open sources, and related literature, and the buffer zone was obtained from the literature review as shown in Table 2. The restriction layers are unified into one layer by build module in ArcGIS as shown in Fig. 5.

The total area of the existing and proposed restriction lands is equal to 11.50% of the country's total area (see Fig. 6). Then, it is reclassified into the binary scale (0 and 1), where all excluded areas that received a value of zero keep that value and indicated that the project is impossible, while areas that obtained a value of one indicated that the project is possible.

Criteria for site selection

GIS data sets of roads, railways, and DEM were collected for the Algeria from different sources, such as National mapping and remote sensing Institute (INCT), open street map (OSM), and The CGIAR Consortium for Spatial Information (CGIAR-CSI). They were summarized as shown in Table 1. The other criteria were calculated such as, hydrogen demand for each commune and potential of hydrogen production from solar energy. For power lines, and slope were drawn and calculated by using ArcGIS software.

Roads

In this study, we consider only the national roads. The data are collected from National mapping and remote sensing Institute [77] and OSM, modified and updated by using Google Earth Pro (see Fig. 7). The minimum distance is required for operation and maintenance purpose reduce construction costs for new access roads [71]. Hydrogen

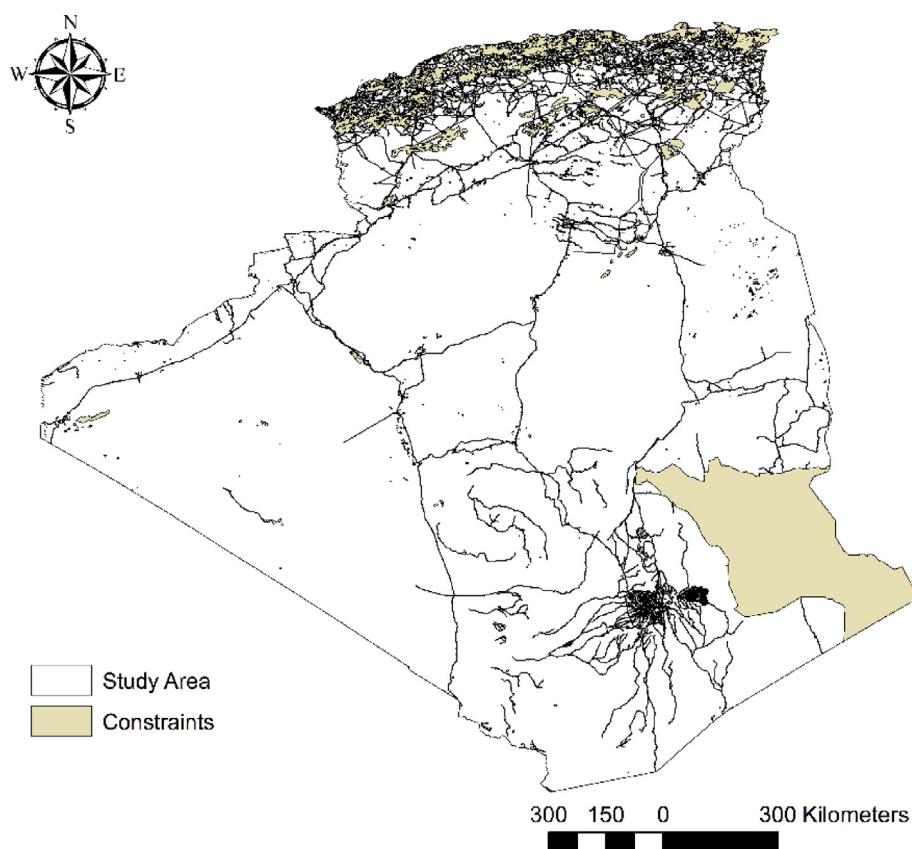
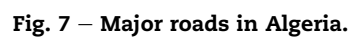


Fig. 6 – Restrictions layer map.



The distance to rail network is a crucial factor to emphasize transferring of passengers and goods. The sites for renewable hydrogen production should be as near as possible as to rail network, a 500 m buffer zone a way of railways was made in

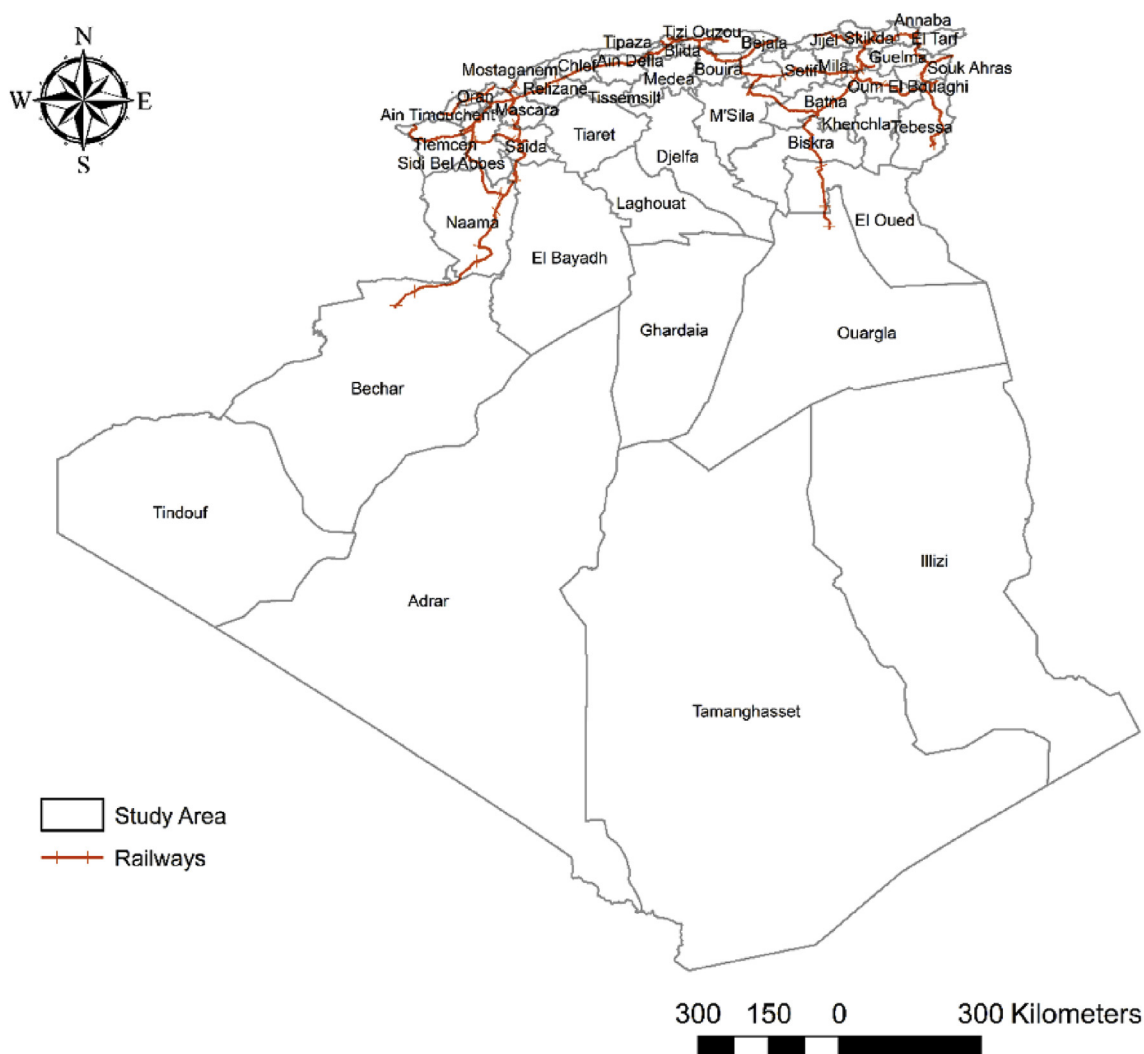


Fig. 8 – Railways in Algeria.

terms of infrastructure [55]. The closer the distance to railways the higher the suitability score. The data was collected from national mapping and remote sensing Institute [77] and OSM, modified and updated by using Google Earth Pro (see Fig. 8).

Power lines

The proximity to power lines connection is necessary to deliver the excess energy from photovoltaic panels to the grid and supply stable power to the other electrical equipment in the system when the solar energy is not enough. A buffer zone of 250 m away from power lines is needed in terms of infrastructure [55]. The closer the distance to grids the higher the suitability score. The data was drawn by using ArcGIS software (see Fig. 9).

Digital elevation model (DEM)

The height criterion was evaluated using a digital elevation model with a 90×90 m resolution obtained from CGIAR-CSI [78]. Global DEM indicates that the terrain heights of Algeria range from -73 m to 2864 m with an average of 554.908 m

as shown in Fig. 10. The area that has higher altitude receive the higher the suitability score since the temperature level in the high range is much better and within the most suitable range for PV panels and hence, the efficiency is definitely higher than the normal conditions.

Slope

The slope of the land surface is a crucial factor as far as construction costs are concerned, because steep slopes of a surface can reduce the accessibility of trucks and increase building costs. Flat terrain is essential for large-scale PV farms; as such, high slope areas are not preferable for such projects due to low economic feasibility [38]. The slope of the land surface was calculated by using ArcGIS software based on the raster map of Algeria's DEM as shown in Fig. 10. Again, lower slopes are preferred and receive higher value scores (see Fig. 11).

Estimating hydrogen demand distribution for each commune
Modeling the hydrogen demand. The determination of sites for hydrogen production in Algeria by using solar energy is an

important issue in understanding a transition to a hydrogen economy. In this work, the estimation of hydrogen demand distribution in the Algeria road transport for each commune is analyzed. We use a model developed in ArcGIS and is mentioned in Ref. [79], which is based on spatial characteristics, namely: population, per-capita vehicle ownership, projections for daily hydrogen use per vehicle, and market penetration level. The following describes the data and factors we used in our analysis:

- The population data and the annual growth rate are provided by the national office of statistics [80] and used in project calculation. The number of total population at time t of project was calculated using the following equation:

$$P_p = P_i \times (1 + \alpha)^{p-i} \quad (1)$$

With P_p the future population value, P_i is the initial or actual population, and α is the annual growth rate.

- Vehicle fleet are taken from the Algerian National Office of Statistics [81]. For projection to the years (2018–2048), we

used the equation one with annual average growth rate factors for vehicle fleet for each region equal to 4.4%.

- The rate of ownership (number of vehicles per person) is calculated for each commune considering all road transport vehicle types (Light Duty Vehicle 63.38%, Light Duty Truck 19.78%, Heavy Duty Truck 7.9%, Agricultural Tractor 2.85%, Trailer 2.65%, Bus 1.59%, and Others 1.85%).
- The annual consumption of vehicles is calculated by assuming that the average annual mileage driven by a vehicle is 38,000 km [82], and consumes one kg of hydrogen every 80 km.
- The temporal aspect of the hydrogen demands is accounted by considering an increasing hydrogen vehicle market share with time (7, 17.9, 32.6, 52.1, 73.7 and 100).

Based on the data, the equation for calculating annual hydrogen demand is given as:

$$AD_{H_2} = P_d \times V_d \times M_p \times H_u \quad (2)$$

where AD_{H_2} is the annual hydrogen demand (kg H_2 /year) for each commune, P_d is the inhabitant, V_d is the per-capita vehicle ownership in each commune (Veh./inhabitant), M_p is the hydrogen vehicle market penetration rate, and H_u is the

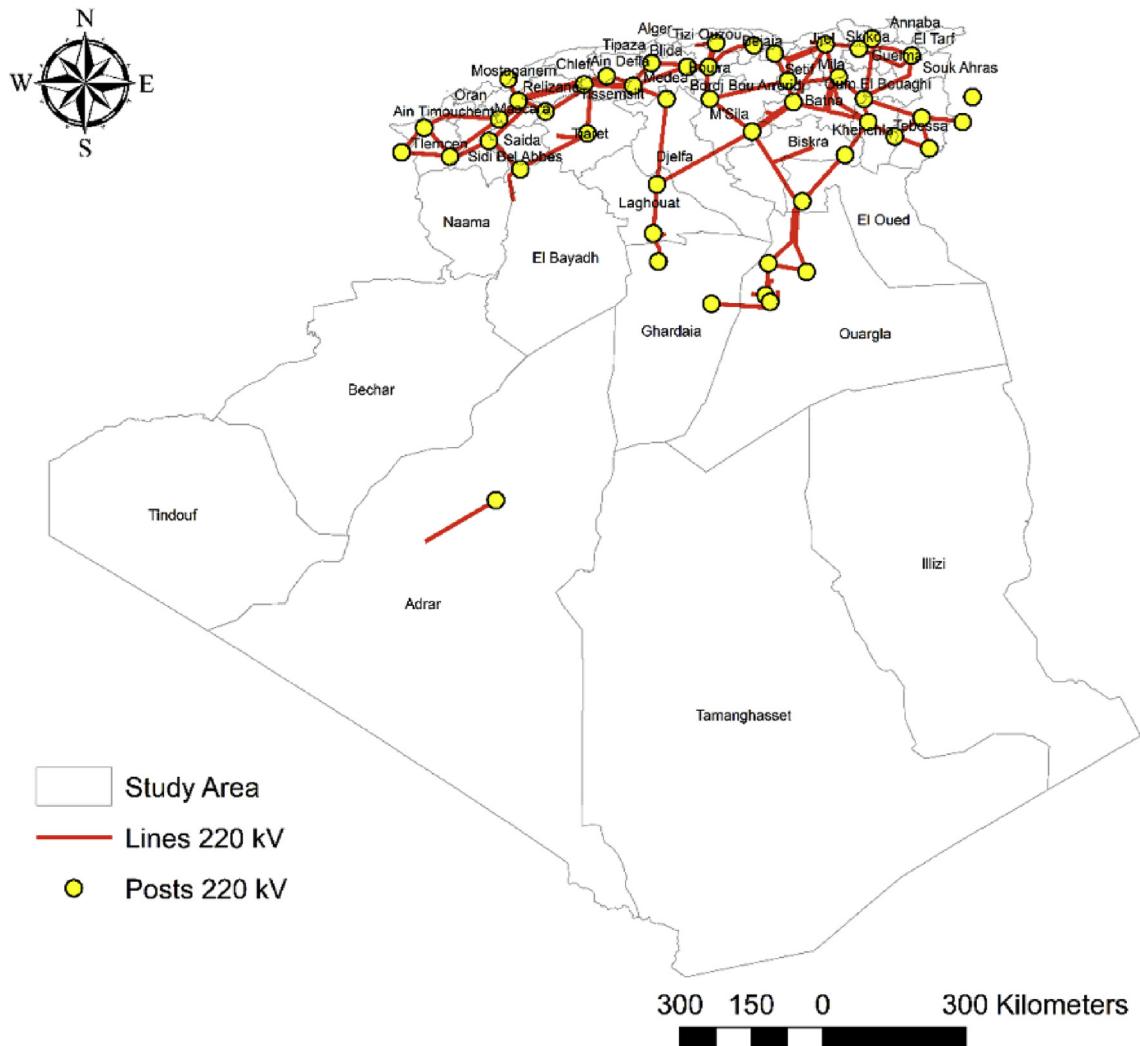


Fig. 9 – Power lines in Algeria.

projected average annual hydrogen use per vehicle (483 kg H_2 /hydrogen vehicle/year) (see Fig. 12).

Potential to produce solar hydrogen

In Algeria, the estimation of hydrogen amount produced by renewable energies mainly depends on the estimation of solar energy potential. Excel spreadsheets were used in order to assess the potential of hydrogen production using mathematical models. Starting by modeling the annual energy produced by the photovoltaic panel which can be expressed as follows [83]:

$$E_{PV} = \eta_{PV} * \eta_{PG} * G \quad (3)$$

G Annual horizontal solar irradiation, $kWh/m^2/year$

η_{PV} Module reference efficiency, %

η_{PG} Power conditioning efficiency, %

To estimate the solar electricity production, the mono-crystalline photovoltaic panel was selected with a power of 250 W at peak, with an efficiency of 15.28% [84].

the proposed system of hydrogen production from solar energy consists of a PV panels, AC/DC converter and Polymer Electrolyte Membrane (PEM) electrolyzer. We consider an electrolytic production rate of 52.5 kWh/kg of hydrogen which is equivalent to about 75% in efficiency. The energy transferred to electrolyzer is demonstrated by the following equation [83]:

$$E_{H_2} = \eta_1 * \eta_2 * E_{PV} \quad (4)$$

η_1 Electrolyzer operation efficiency, %

η_2 Electrolyzer losses, %

E_{PV} Solar electric source production, $kWh/km^2/year$

The annual mean of solar hydrogen potential is illustrated as follows [85]:

$$M_{H_2} = \frac{E_{H_2}}{HHV_{H_2}} = \frac{\eta_{Elec} E_{PV}}{HHV_{H_2}} \quad (5)$$

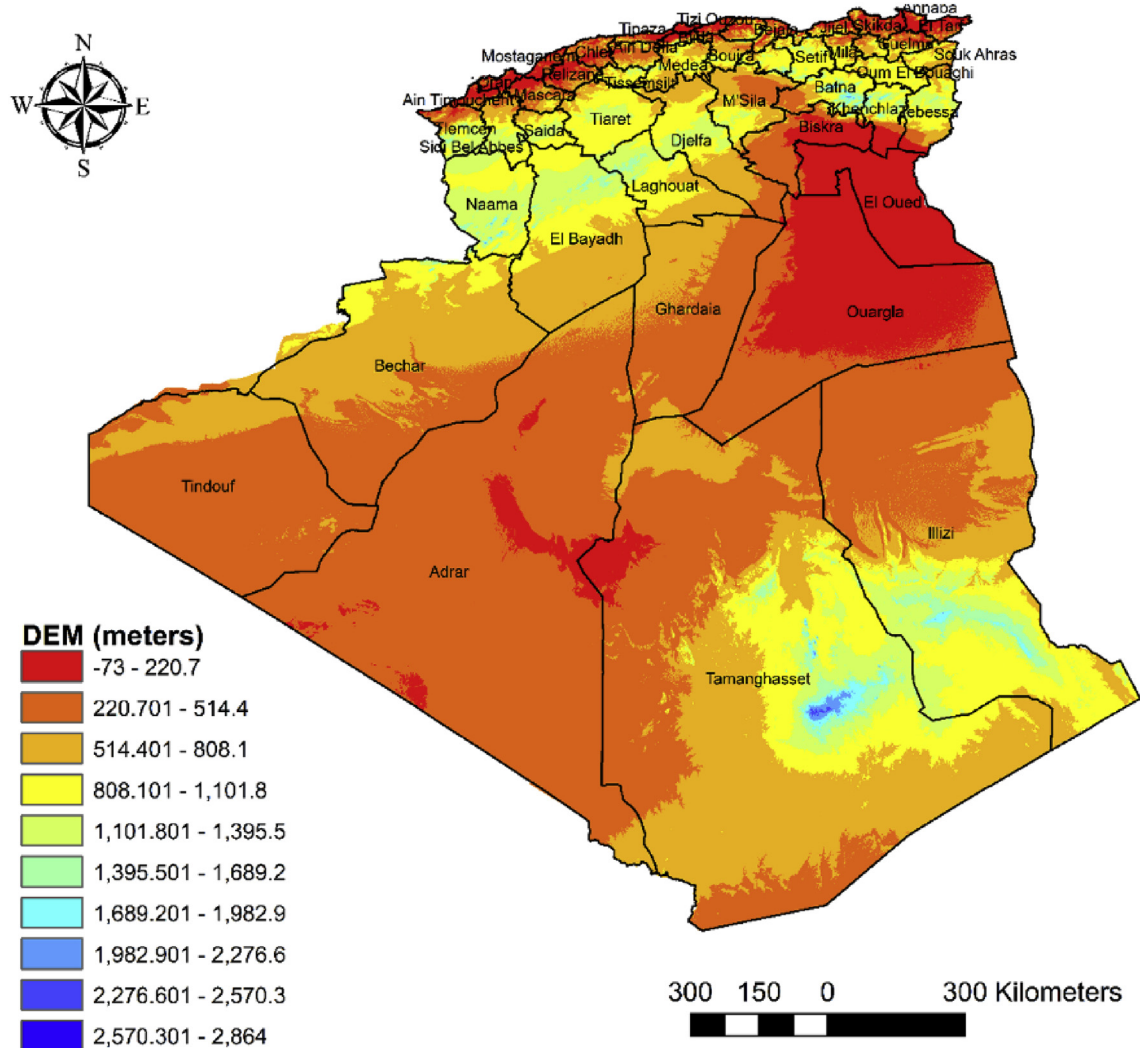


Fig. 10 – Digital elevation model in Algeria.

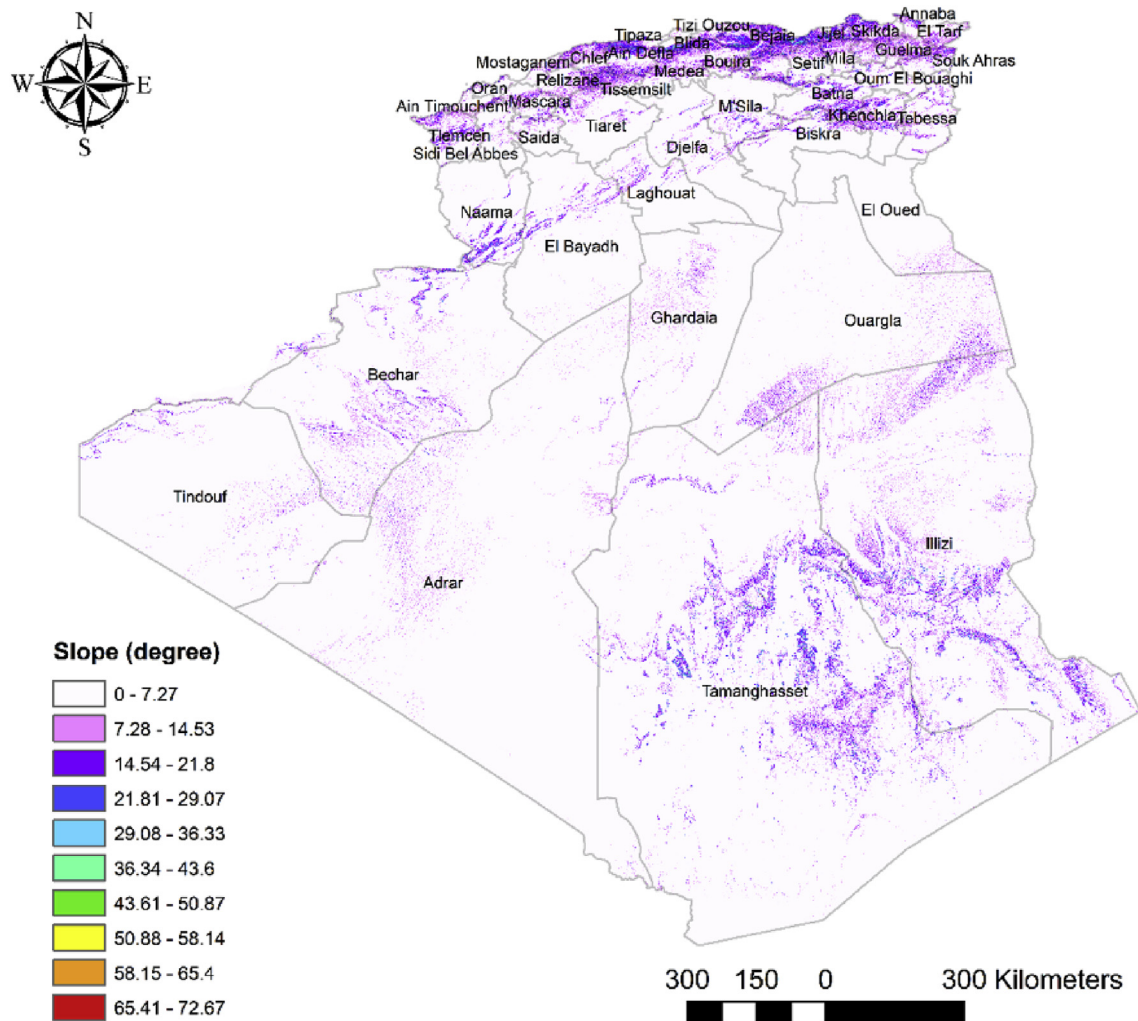


Fig. 11 – Slope in Algeria.

M_{H_2} is the annual hydrogen production [$\text{kg}/\text{km}^2/\text{year}$];
 E_{H_2} is the hydrogen energy produced [$\text{kWh}/\text{km}^2/\text{year}$];
 HHV_{H_2} is the hydrogen higher heating value ($39.4 \text{ kWh}/\text{kg}$);
 η_{Elec} is the efficiency of the electrolysis system.

The area that has more potential to produce hydrogen from solar energy is preferred and receive higher value scores as shown in Fig. 13.

Using GIS-AHP based approach to select suitable sites for renewable hydrogen production

The selection of the most suitable sites to install solar-powered hydrogen production system is a very complex issue. It requires the consideration of multiple alternative solutions and evaluation criteria. Usually to overcome this type of problem researchers and decision makers use the Multi-Criteria Decision Making (MCDM) methods [39,43,68,71,74,86–89]. MCDM includes several different methods, [39] presented a literature review of more than 90 published papers on multi-criteria decision making on sustainable energy planning and observed that The analytic hierarchy process (AHP) is the most popular technique [38,47,66,90].

The hierarchy of locating renewable hydrogen production installation system was establishing and Fig. 3, presents the adopted criteria, that can be divided generally into two groups. The first group includes economical criteria and the second group is the technical criteria. AHP is a systematic approach developed in the 1970s by Thomas Saaty to give decision-making based on experience, intuition, and heuristics. In order to apply AHP, eight steps have to be conducted consecutively [91,92] as described below:

Step 1: A structural hierarchy is formed for the decision-making process consists of several levels to establish the goal which is selection suitable sites for renewable hydrogen production installation system, two criteria were created, economical and technical. These criteria were then divided into four and three sub-criteria respectively as shown in Fig. 3.

Step 2: Data are collected from experts corresponding to the hierarchical structure, in this step the criteria are compared with each other based on our goal, while all the other sub-criteria are compared with each other based on each criterion. Different scales and their meanings are explained in Table 3.

Step 3: The pair-wise comparison of criteria/sub-criteria i with criteria/sub-criteria j yields a square matrix $A_{n \times n}$ where a_{ij} denotes the comparative importance of criteria/sub-criteria i with respect to criteria/sub-criteria j . In the matrix, $a_{ij} = 1$ when $i = j$ and $a_{ij} = 1/a_{ji}$.

criteria / sub – criteria $A_1 \ A_2 \ A_3 \dots \dots A_n$

$$A_{n \times n} = \begin{bmatrix} A_1 & 1 & a_{12} & a_{13} & \dots & \dots & a_{1n} \\ A_2 & a_{21} & 1 & a_{23} & \dots & \dots & a_{2n} \\ A_3 & a_{31} & a_{32} & 1 & \dots & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \ddots & \vdots \\ A_n & a_{n1} & a_{n2} & a_{n3} & \dots & \dots & 1 \end{bmatrix}$$

Step 4: In this step, we find the relative normalized weight (w_i) of each criteria/sub-criteria by calculating the geometric mean of the i -th row, and normalizing the geometric means of rows in the comparison matrix.

$$GM_i = \left[\prod_{j=1}^n a_{ij} \right]^{1/n} \quad (6)$$

$$w_i = \frac{GM_i}{\sum_{i=1}^n GM_i} \quad (7)$$

Step 5: Determine the maximum Eigenvalue λ_{max} by product the original matrix and the transpose of the normalized weights, and then the resultant is divided by the normalized weights and average the final matrix to get the maximum Eigenvalue.

Step 6: [93] has proposed a consistency index (CI), which is related to the eigenvalue method. The consistency of the matrix of order N is evaluated. If this consistency index failed to reach a threshold level, then the answers to comparisons were re-examined.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (8)$$

Where n is dimension of the matrix and λ_{max} is maximal eigenvalue. This CI can be compared with that of a random matrix, RI.

$$CR = \frac{CI}{RI} \quad (9)$$

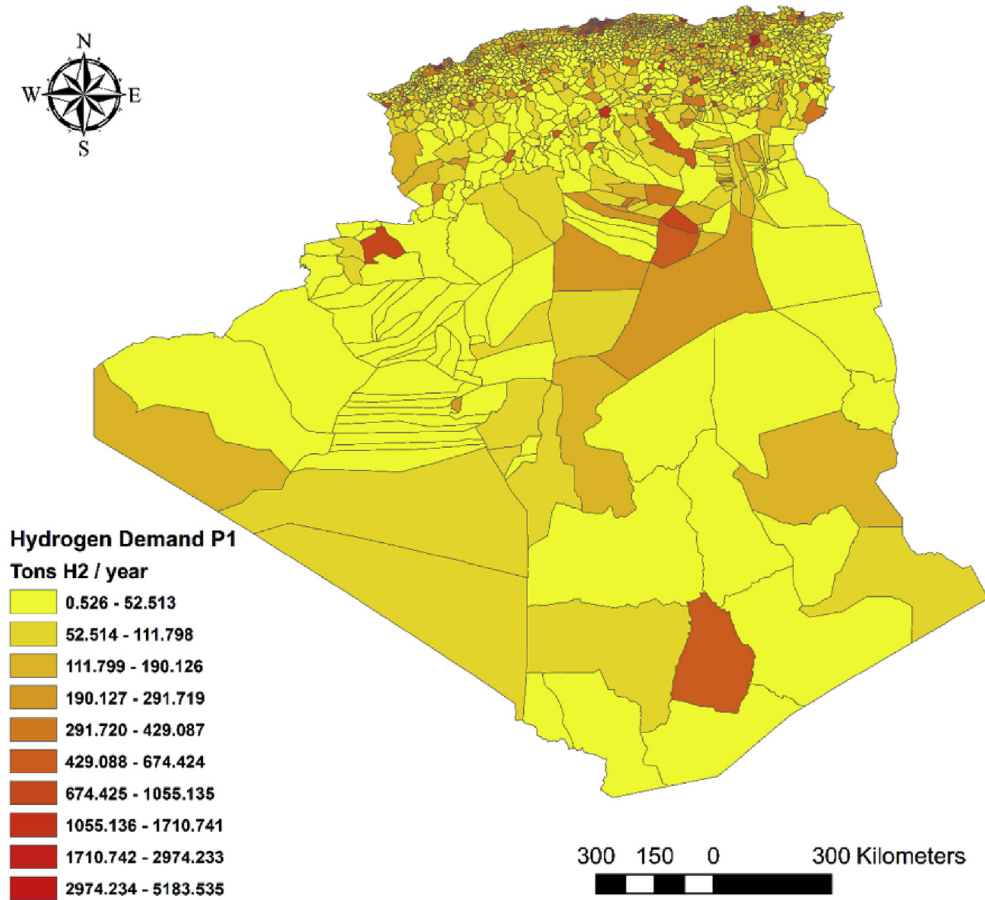


Fig. 12 – Spatial distribution of annual hydrogen demand by commune.

Where CR is the consistency ratio and RI is the random index (the average CI of 500 randomly filled matrices) as shown in Table 4. Saaty suggests the value of CR should be less than 0.1.

Step 7: In this step priorities of each criterion with respect to goal is multiplied with the priority of each sub-criteria with respect to that criterion. The sub-criteria with the higher value will be the preferred sub-criteria.

Step 8: For Aggregation of individual judgments (Consolidation of participants), we use the weighted geometric mean of the decision matrices elements $a_{ij}(k)$ using the individual decision maker's weight w_k as given in the equation below [94]:

$$c_{ij} = \exp \left(\frac{\sum_{k=1}^N w_k \ln a_{ij}(k)}{\sum_{k=1}^N w_k} \right) \quad (10)$$

where c_{ij} is consolidated decision matrix, k participant, and N number of participant. In our case, all individuals receive equal weights.

Finally, after the assignment of the criteria's value scores that were obtained from the AHP tool. Using the Raster

Calculator tool in the ArcGIS, the selection suitable sites for renewable hydrogen production installation system is tackled as follows:

1. Calculate the distance ranges of distance-dependent criteria using the EUCLIDEAN DISTANCE tool in ArcGIS. Then, the RECLASSIFY tool is employed to assign the value score to the criteria into a common preference scale of suitability ranging from 1 to 10 (with 10 being the most suitable) as shown in Table 5.
2. We applied the RASTER CALCULATOR, which enables the construction and execution of a single map algebra expression in Python syntax, which combines the assigned value scores for each criterion with their relative importance according to the AHP. By doing so, we created the weighted overlay function in order to combine the assigned value scores for each sub-criterion with their relative importance to generate the ultimate combined layer across the study area.
3. Lastly, to obtain the land suitability index (LSI), Eq. (11) has been applied for each pixel of the study area layer. If restriction (r) exists, then $r = 0$ which leads to the LSI value of an unsuitable site.

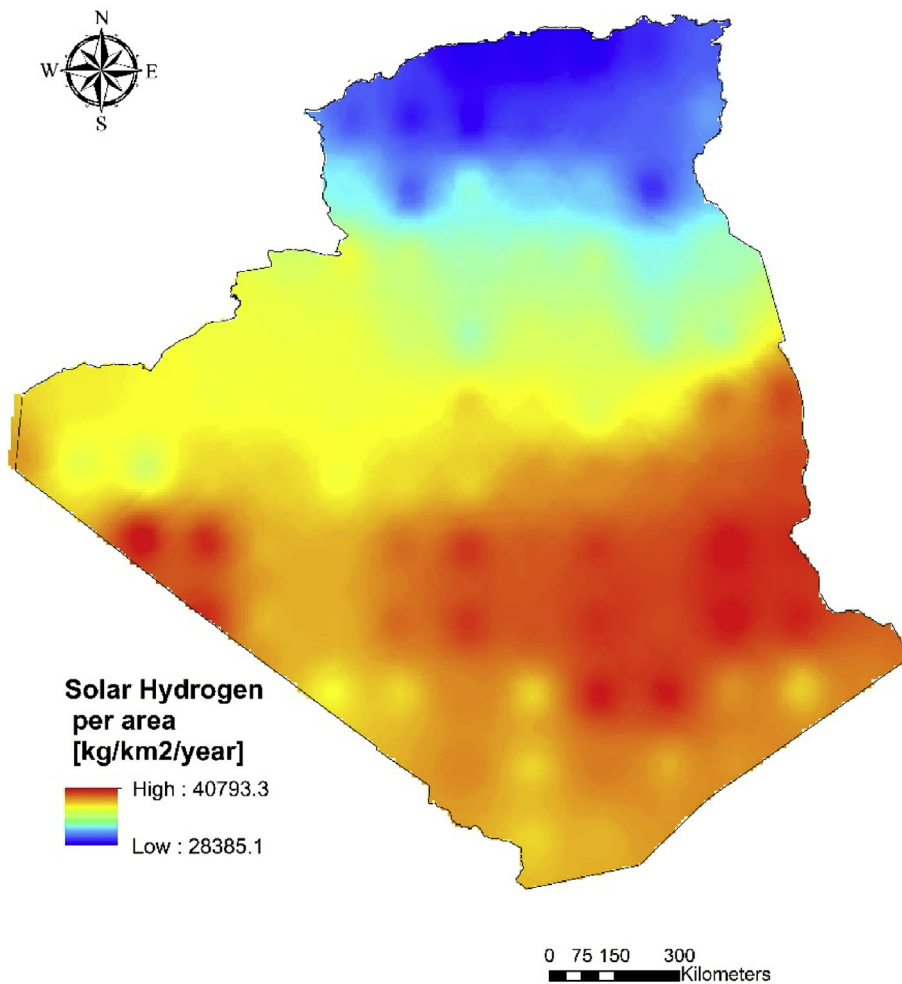


Fig. 13 – Hydrogen potential from solar energy.

Table 3 – The comparison scale in AHP [90].

Intensity of importance	Definition	Explanation
1	Equal importance	Two activities contribute equally to the objective
3	Weak importance of one over another	Experience and judgment slightly favor one activity over another
5	Essential or strong importance	Experience and judgment strongly favor one activity over another
7	Demonstrated importance	An activity is strongly favored and its dominance is demonstrated in practice
9	Absolute importance	The evidence favoring one activity over another is of the highest possible order of affirmation
2, 4, 6, 8	Intermediate values between the two adjacent judgments	When compromise is needed

$$LSI = \sum_i^n x_i \cdot w_i \cdot r \quad (11)$$

where $r \in \{0, 1\}$, x_i is criteria value, and w_i is criteria weight.

In this paper, ArcMap 10.2 was utilized to perform spatial processes and manipulation for both vector and/or raster files of the study area's dataset.

Results and discussion

To determine the suitable areas for solar hydrogen production installation system, AHP was used to assess and evaluate scores based on suitable criteria. Each criterion map and constraints map were prepared using ArcGIS.

Comparison of the criteria

The comparison of the criterion is carried out in excel spreadsheet as shown in Tables 6–8.

Table 6 illustrates the pairwise comparison consolidated decision matrix respect to select suitable sites for solar hydrogen production. We notice that economical criteria has the high weight with the value of (11/5) compared to technical criteria with the value of (5/6) according to experts' opinions.

Table 7 shows the pairwise comparison consolidated decision matrix respect to economical criteria. From the Table 7 it shows that the proximity to roads has the higher value of weight among other sub-criteria, followed by the sub-criteria of proximity to power lines. Lastly, the proximity to the railways has the low value (2/5) compared to roads and to power lines.

Table 8 illustrates the pairwise comparison consolidated decision matrix respect to technical criteria. According to Table 8 the hydrogen demand has a higher value compared to potential hydrogen production with (24/9), to DEM with a value of (48/9), to slope (54/7), followed by potential hydrogen production, DEM, and slope.

Table 9 presents the weight of criteria with respect to the goal and the weight of sub-criteria with respect to criteria obtained by pairwise comparisons of criteria and sub-criteria. The economical criteria is important with a value of 0.543 then the technical criteria 0.457 and proximity to roads is the important sub-criteria with 0.265, 0.233, 0.187, 0.157, 0.091, 0.040, and 0.027 for hydrogen demand, proximity to power lines, potential renewable hydrogen production, proximity to railways, DEM, and slope, respectively.

Suitable sites for renewable hydrogen production

The final locations were defined by overlaying the results of the restrictive and the different weighted criteria, which allows us to classify the study area on a scale between 0 and 6.5588 where the pixel that has a high value corresponds to

Table 4 – Random consistency indexes [93].

n	1	2	3	4	5	6	7	8	9
RI	0.00	0.00	0.58	0.90	1.12	1.24	1.32	1.41	1.49

Table 5 – AHP criteria and value scores.

Value score	Potential of hydrogen production (Tons/km ² /year)	Hydrogen demand (Tons H ₂ /year)						Distance from road network (m)	Distance from railways (m)	Distance from transmission lines (m)	Height (m)	Slope (degree)
		P ₁	P ₂	P ₃	P ₄	P ₅	P ₆					
1	29625.94	102.15	304.77	424.24	545.21	1283.96	8554.51	>5000	>5000	>2500	- 73	>5.14
2	30866.76	346.06	988.75	1906.23	3267.57	6417.37	22811.23	4500–5000	4500–5000	2250–2500	200	4.57–5.14
3	32107.58	589.97	1748.73	4023.37	7623.35	15400.83	34216.61	4000–4500	4000–4500	2000–2250	400	3.99–4.57
4	33348.40	752.57	2432.72	5928.79	13612.54	28234.35	48473.33	3500–4000	3500–4000	1750–2000	600	3.43–3.99
5	34589.22	894.85	3344.70	7622.49	20690.69	44917.93	68432.75	3000–3500	3000–3500	1500–1750	800	2.85–3.43
6	35830.04	1098.10	4256.68	10798.2	27224.35	57751.45	85540.82	2500–3000	2500–3000	1250–1500	1000	2.28–2.85
7	37070.86	1606.24	5244.66	15455.89	34302.5	70584.97	139716.36	2000–2500	2000–2500	1000–1250	1200	1.71–2.28
8	38311.69	2622.52	7980.60	21595.58	65337.42	91118.60	202445.95	1500–2000	1500–2000	750–1000	1400	1.14–1.71
9	39552.51	3638.79	11704.51	31546.11	110528.64	204053.58	496134.45	1000–1500	1000–1500	500–750	1600	0.57–1.14
10	40793.33	5183.53	19380.34	53987.73	138841.20	327255.37	727093.37	500–1000	500–1000	250–500	1800	0–0.57

Table 6 – Pairwise comparison consolidated decision matrix for suitable sites for renewable hydrogen production.

Suitable sites for renewable hydrogen production	Economical criteria	Technical criteria
Economical criteria	1	11/5
Technical criteria	5/6	1

Table 7 – Pairwise comparison consolidated decision matrix respect to economical criteria.

Economical criteria	Proximity to Roads (C1)	Proximity to Railways (C2)	Proximity to Power Lines (C3)
Proximity to Roads (C1)	1	24/9	12/3
Proximity to Railways (C2)	2/5	1	2/5
Proximity to Power Lines (C3)	3/5	24/9	1

the most suitable area for hydrogen production based on solar energy as shown in Fig. 14. We notice that the south of Algeria has highly suitable sites (red to yellow colors in the map) and some sites scattered along the map in High Plains according to criteria chosen.

Table 10 presents the final suitability results that were divided into four discrete categories: very low suitable areas, low suitable areas, moderately suitable areas, and highly suitable areas with a manual interval classification method, which is used to define the degree to which site is suitable for renewable hydrogen production according to the associated criteria and excluding all restrictions.

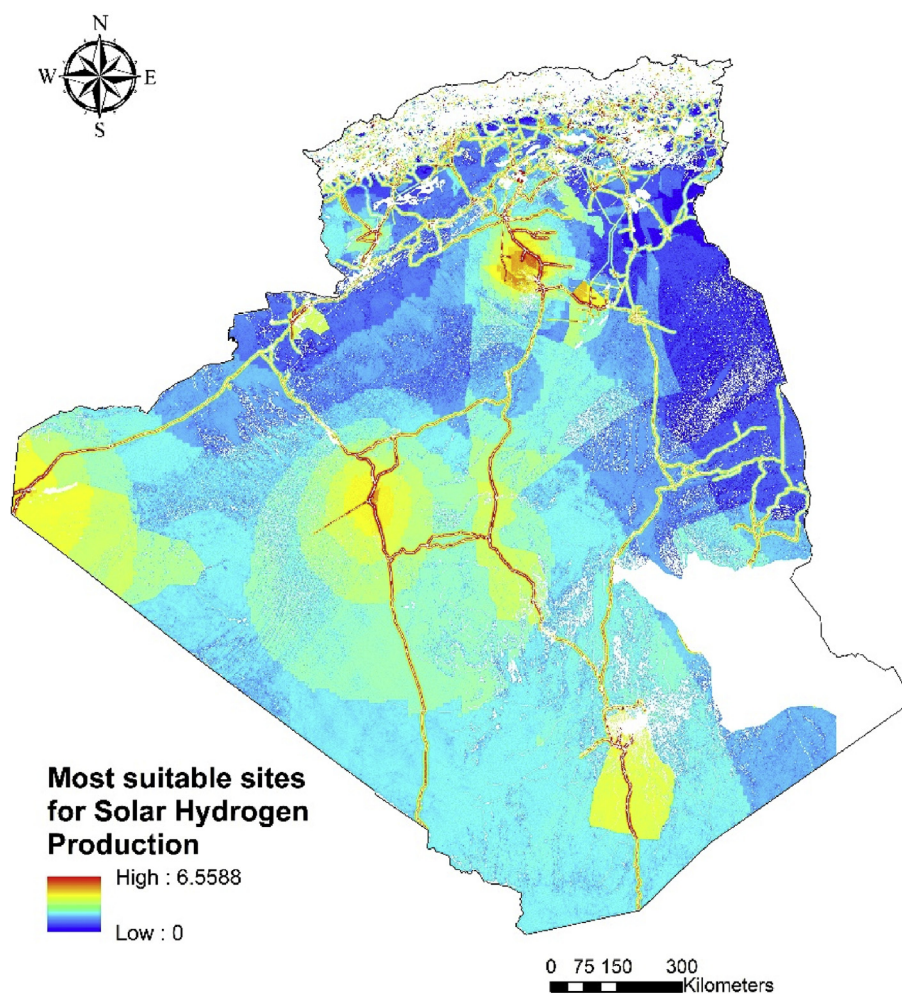
The results indicate that 10.34% (246,272.02 km²), of the study area has very low suitability, 60.75% (1,446,907.65 km²) has low suitability, 6.68% (159,100.3 km²) has moderate suitability and 0.49% (11,669.21 km²) has high suitability for a solar-powered hydrogen production installation system. The other 21.74% (517,790.5 km²) of the study area is not suitable for such projects as shown in Fig. 15. The highly suitable areas (red color in the map) are located in zones that are close to roads due to proximity to roads combined with proximity to power lines and higher potential solar hydrogen production in that region. Along the southeast and High Plains, lands have

Table 8 – Pairwise comparison consolidated decision matrix respect to technical criteria.

Technical criteria	Hydrogen Demand (C4)	Potential Hydrogen Production (C5)	DEM (C6)	Slope (C7)
Hydrogen Demand (C4)	1	24/9	48/9	54/7
Potential Hydrogen Production (C5)	2/5	1	53/8	65/8
DEM (C6)	1/5	1/5	1	16/7
Slope (C7)	1/6	1/7	1/2	1

Table 9 – Significance weights of main criteria, criteria and sub criteria used in selection suitable sites for renewable hydrogen production installation system.

Criteria	Weight	CR	Sub-Criteria	Weight	CR	w_i
Economical criteria	0.543	0	Proximity to Roads	0.488	0.031	0.265
			Proximity to Railways	0.167		0.091
			Proximity to Power Lines	0.345		0.187
Technical criteria	0.457		Hydrogen Demand	0.509	0.059	0.233
			Potential Renewable Hydrogen Production	0.343		0.157
			DEM	0.088		0.040
			Slope	0.06		0.027

**Fig. 14 – The most suitable sites for hydrogen production from solar energy.**

very low suitable LSIs (blue color in map) due to low hydrogen demand and major steep slopes, including the Great Eastern ergs which are the most difficult of all Saharan areas and are

Table 10 – Suitability index.

Suitability	Value score
Unsuitable	0
Very Low Suitable	0–2
Low Suitable	2.01–3
Moderately Suitable	3.01–4
Highly Suitable	4.01–6.9657

generally avoided by modern trans-Saharan routes and the Saharan Atlas which is a series of subranges includes the Ksour Range in the west, the Amour Range in its central and the Ouled-Nail Range at its eastern end. It also includes the Aurès (Belezma), the Hodna Mountains, the Nememcha Range and the Zab Mountains. The south region of the study area illustrates low to moderately LSI (yellow color in the map), since it has higher potential renewable hydrogen production. The results of the AHP method confirm some previously studies which have shown that the south regions are the most favorable sites for hydrogen production via solar energy [15,95–97].

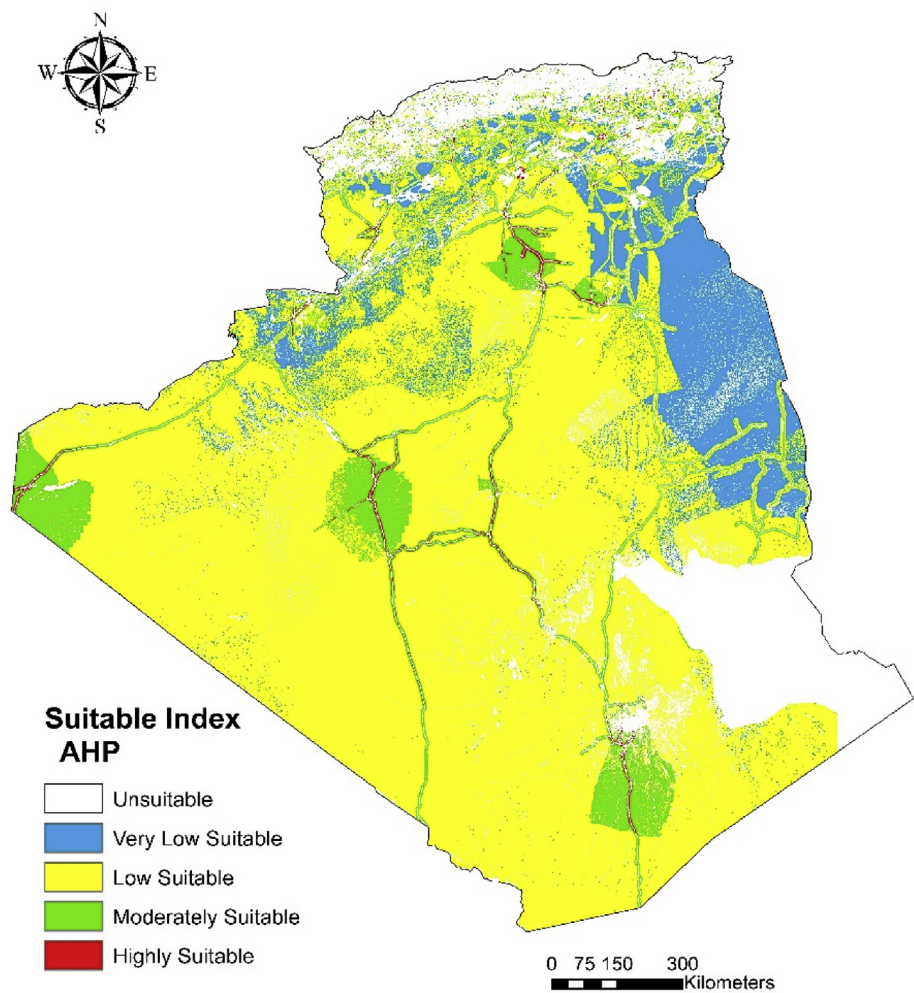


Fig. 15 – The sites suitability map for the solar-powered hydrogen production installation system.

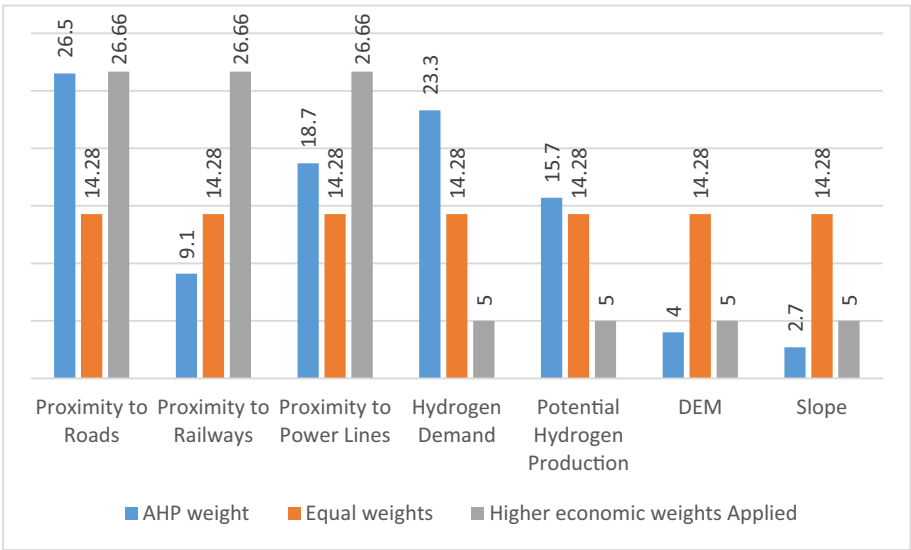
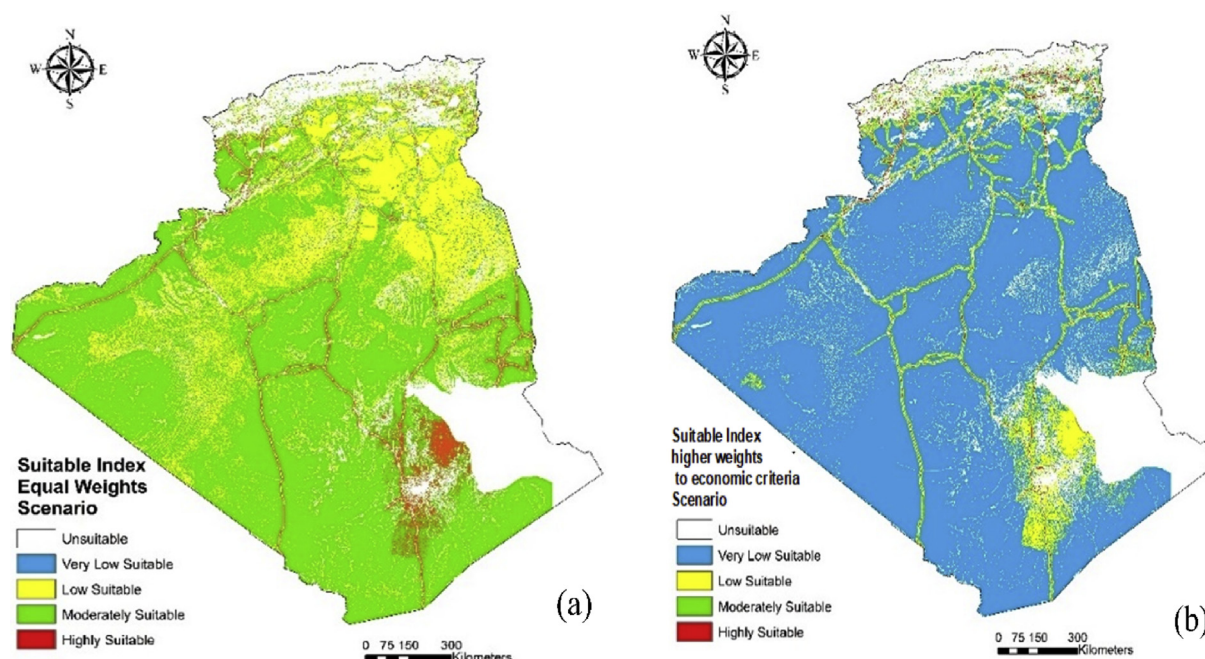


Fig. 16 – Weights of decision criteria considering different scenarios.

Table 11 – Land suitability distribution considering different scenarios with respect to study area.

Scenario	Weights	Land suitability distribution (%)				
		High Suitable	Moderate Suitable	Low Suitable	Very Low Suitable	Unsuitable
AHP	Tech. = 0.457 Eco. = 0.543	0.49	6.68	60.75	10.34	21.74
Equal weights	Tech. = 0.5 Eco. = 0.5	3.75	54.34	19.26	0.90	
Higher economic weights	Tech. = 0.2 Eco. = 0.8	1.05	2.76	5.15	69.29	

**Fig. 17 – Land suitability results considering equal weights criteria (a) and higher weights to economic criteria (b).**

Sensitivity analysis

In a multi-criteria decision making a “what if”, sensitivity analysis is recommended as a means of checking the stability of the results against the subjectivity of the expert judgments. By doing so, different criteria weight scenarios were considered and their overall impact on land suitability index was assessed. In this way, two scenarios including equal weights [28,45,49] and higher economic weight [38] have been examined in this study as shown in Fig. 16.

In the first scenario, all criteria have the same weights, the weight of 14.28% has been assigned to each criterion. The results obtained indicate a comparable distribution of suitability across the study area, where most areas are characterized by moderate suitability (green color in the map) with a value of 54.34%. However, there are considerable shifts across the different value scores and suitability classes. Compared to AHP methodology, very low suitable areas (blue color in the map) and low suitable areas (yellow color in the map) have decreased from 10.34% to 0.90% and from 60.75% to 19.26% respectively around the whole study area, and the high suitable area (red color in the map) showed an increase from

0.49% to 3.75% of the study area (see Table 11). Most of the moderate to high suitable areas are spread in the Sahara and slightly in the High Plains of Algeria, as depicted in Fig. 17 (a). More moderate and high LSI sites exist where proximity to roads and hydrogen demand weights have decreased from 26.5% to 14.28% and from 23.3% to 14.28% respectively. The very low suitable and low suitable LSI improved to moderate due to technical criteria weights.

In the second scenario, the economic criteria including proximity to roads, to railways, and to power lines have been assigned the higher value among other criteria. The results show that most study area is characterized by the very low suitable area (blue color in the map) which is six times superior compared to AHP method and seventy-six times to equal weights where there is no infrastructure in these areas as shown in Fig. 17 (b).

The results of different scenarios have proven sensitivity to the criteria weights and offer various land suitability distribution. Table 11 shows the final results obtained by varying the criteria weights, thus demonstrating that both technical and economic criteria are influential in the evaluation of the study area.

Discussion

This study has both direct and indirect benefits to Algeria. The results will directly help the Algerian authority to recognize suitable locations for the renewable hydrogen production installation system and to help promote the implementation of hydrogen energy in Algeria. The results, not only indicate where suitable sites are located, but also show that not all of Algeria's land territory is suitable to locate hydrogen production installation system based on solar energy.

The main limitation of this work is that the size and amount of hydrogen production installation system were not taken into account. This can be addressed by facility location optimization models that are able to consider the size and amount of hydrogen needed to be produced on each installation system.

The work developed in this paper allows such a location model to be much easier to solve and also it can be applied to investigate other types of similar projects in Algeria.

Conclusion and future works

The selection of locations for renewable hydrogen production installation system is a complex process due to the different security, economic, environmental, technical, and social requirements that must be considered. Locations with the highest hydrogen demand or potential of renewable hydrogen production cannot always be selected and several other criteria play significant roles in selecting convenient locations. Therefore, the use of MCDM models becomes necessary.

This paper presents an application of combining MCDM with GIS for sites selection of solar hydrogen production installation system in Algeria. The objective of the study was to find suitable sites to host hydrogen production installation system from solar energy resources taking into account a number of different criteria. AHP was utilized to assign the relative weights of the evaluation criteria, while GIS established the spatial dimension of constraints and evaluation criteria and elaborated them in order to produce the overall suitability map. Furthermore, by incorporating associated criteria into the decision making process makes suitable sites for solar hydrogen production installation system project more economically and technically feasible. Technical and economic criteria considered in this study. Each of the mentioned criteria has a number of sub-criteria. Hydrogen demand, potential solar hydrogen production, DEM, and slope for technical criteria and proximity to roads, railways, and power lines for economic criteria.

As a result, the final index model was grouped into four categories as “very low suitability”, “low suitability”, “moderate suitability” and “high suitability” with a manual interval classification method. The results indicate that 10.34% (246,272.02 km²), of the study area has very low suitability, 60.75% (1,446,907.65 km²) has low suitability, 6.68% (159,100.3 km²) has moderate suitability and 0.49% (11,669.21 km²) has high suitability for a solar-powered hydrogen production installation system. The other 21.74% (517,790.5 km²) of the study area is not suitable for such projects.

In a multi-criteria decision making a “what if”, sensitivity analysis is recommended as a means of checking the stability of the results against the subjectivity of the expert judgments. Two scenarios including equal weights and higher economic weight have been examined in this study. The sensitivity analysis highlights that the suitable sites for solar hydrogen production installation system are dependent on the weights of the criteria that influence the decision.

The main advantage of this work is exploiting the existing resources and infrastructure to provide feasible sites to implement renewable hydrogen production installation system. However, our results describe for the first time the suitable sites to produce solar hydrogen in Algeria using MCDM methods coupled with GIS.

In future research, tackling more than two supply criterion, such as wind, biomass and geothermal. Moreover, applying a new approach as well as performing a comparative analysis of such techniques to determine the most suitable approach for future research.

The MCDM methodology integrated with GIS is a powerful tool for effective evaluation of the solar-powered hydrogen production sites selection.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ijhydene.2019.10.099>.

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